

PetroPro Modernization Master Prompt

Purpose

You are an expert software architect and senior full-stack engineer. I have attached the complete legacy FoxPro 2.6 DOS application as a ZIP containing PRG, DBF, FRX/FRT, indexes, menus, configuration files and sample data. Your objective is NOT to perform a literal code conversion. Reverse engineer the application, preserve every business rule, and rebuild it as a modern offline-first Progressive Web Application (PWA) named **PetroPro**.

Product

Product Name: **PetroPro**

Tagline: **The Complete Petrol Station Management Platform**

Technology Stack

Frontend: Next.js, React, TypeScript, Tailwind CSS, PWA.

Backend: Node.js (Fastify or Express) with REST APIs.

Data: Preserve DBF compatibility initially using a repository layer; allow future migration to SQLite/PostgreSQL.

Authentication: Local users and roles.

Reports: PDF, Excel and print-ready invoices.

Mandatory Deliverables

1. Analyze every source file and document business logic.
2. Produce ER diagram, data dictionary and module documentation.
3. Design a modern responsive UI.
4. Recreate all billing, inventory, reports and workflows.
5. Validate outputs against the FoxPro system.

New Features

- Fuel attendant mobile PWA for Android tablets/phones.
- Attendant enters vehicle number, fuel type, quantity/amount and optional odometer.
- Transaction appears instantly in cashier Pending Queue.
- Cashier selects transaction, adds lubricants/accessories if needed, accepts payment and prints invoice.
- Offline synchronization over local network.
- Dashboards, analytics, user roles, shift management, fleet customers, audit logs and backup/restore.

Coding Standards

Create modular, production-quality code with clear folder structure, reusable components, API documentation, test cases and migration notes. Generate the project incrementally with complete explanations for each phase.

Success Criteria

The new PetroPro application must preserve all existing FoxPro functionality while providing a modern, installable offline PWA that is faster, easier to use and ready for future commercial deployment.